VGAC: A Full-Stack GPU Cluster Observability Platform with Calibration-Gated Predictive Intelligence

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Abstract

Calibrated queue-delay predictions are useful only if they translate into scheduling decisions with known operating characteristics. We present VGAC, a full-stack GPU cluster observability platform with calibration-gated predictive intelligence: each predictive intervention is permitted only when the underlying probability is calibrated well enough to support it. The contribution has three parts. (1) A graduated intervention framework with four tiers—Annotate, Warn, Suggest, Gate—each carrying an explicit calibration prerequisite (Expected Calibration Error $\leq$ 0.10 for Annotate down to ECE $\leq$ 0.03 for Gate) and a tier-specific probability threshold ε_a in the rule "apply action a if $\hat{p} \geq \varepsilon_a$." (2) A submit-time queue-state capture methodology that corrects an inverted feature correlation, taking Pearson r from -0.27 under naive 30 s polling to $+0.44$ under submit-time capture, and so establishes correct measurement as a prerequisite for correct decision rules. (3) An empirical floor-to-ceiling validation on an Amazon EKS GPU cluster: a 2-feature "floor" model qualifies for Annotate and is marginal for Warn (AUROC 0.756, ECE 0.077), while the 17+ feature production VGAC model achieves ECE 0.005 and qualifies for all four tiers including autonomous admission gating. The framework is realized in production-grade Kubernetes admission webhooks and Slurm `job_submit` hooks that re-evaluate tier qualification at serve time so degraded models automatically reduce their action scope.

CCS Concepts

• **Computer systems organization** → **Cloud computing**; • **Computing methodologies** → *Supervised learning by classification.*

Keywords

GPU scheduling, calibration, decision rules, policy integration, Kubernetes, Slurm, observability

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1 Introduction

A growing body of work has demonstrated that queue-delay predictions can be made calibrated—that $\hat{p} = P(\text{delay} > T)$ can be made to match realized frequencies [4]. A companion paper [1] defines the service-level indicators (SLIs) and service-level objectives (SLOs) for monitoring this calibration in production. But a calibrated probability alone does not tell a scheduler what to do. The missing piece is a decision-theoretic bridge: formal rules that translate $\hat{p}$ into actions, with explicit prerequisites on how calibrated $\hat{p}$ must be for each action to be trustworthy.

Current GPU schedulers [6, 7] are reactive: they queue a job and observe what happens. Predictive systems can improve this by providing risk estimates at submit time. But the operational question remains: given $\hat{p} = 0.73$ for a submitted job, should the system (a) do nothing, (b) annotate the job with a warning, (c) suggest rescheduling, or (d) gate the submission pending confirmation?

The answer depends on two factors: the severity of the action and the trustworthiness of $\hat{p}$. A low-stakes annotation can tolerate moderate miscalibration; an admission gate that blocks submissions must be backed by highly calibrated probabilities, or users will lose trust and override the system. We formalize this intuition into a graduated intervention framework and instantiate it in VGAC, a full-stack observability platform whose predictive intelligence is *calibration-gated*: each tier of intervention is permitted only when the underlying model has earned the calibration quality the action requires.

1.1 Contributions

We contribute: (1) a decision-theoretic framework with four intervention tiers, each paired with a formal calibration prerequisite; (2) the decision rule "apply action a if $P(\text{delay} > T) \geq \varepsilon_a$" with tier-specific thresholds; (3) a submit-time queue-state capture methodology that corrects an inverted feature correlation ($r = -0.27 \rightarrow r = +0.44$); (4) an honest tier-qualification analysis showing the 2-feature floor model (ECE 0.077) qualifies for Annotate but not Suggest/Gate, while the 17+ feature production VGAC model (ECE 0.005) qualifies for all four tiers; and (5) reference implementations for Kubernetes mutating admission webhooks and Slurm `job_submit` plugins that enforce tier qualification dynamically at serve time.

2 Background

2.1 From Prediction to Decision

Prior work on cluster-scheduling prediction [3] focuses on placement optimization; our focus is on the human-system interface at

Table 1: Graduated intervention tiers with calibration prerequisites.

Tier	Action	ε_a	ECE Prereq.
1: Annotate	Add risk label	0.3	≤ 0.10
2: Warn	Notify user	0.5	≤ 0.07
3: Suggest	Recommend reschedule	0.7	≤ 0.05
4: Gate	Require confirmation	0.9	≤ 0.03

submit time. Calibration methods [4, 5] ensure $\hat{p}$ matches reality; we address what to do with a calibrated $\hat{p}$.

2.2 Decision Theory for Classification

In medical diagnosis, a calibrated probability $P(\text{disease})$ is combined with a cost matrix to determine whether to treat [8]. We apply the same logic to scheduling: each tier has an action cost (from zero for annotation to high for submission gating), and the calibration prerequisite ensures the expected cost of acting on a miscalibrated $\hat{p}$ remains below an acceptable threshold.

3 Graduated Intervention Framework

3.1 Decision Rule

Let $\hat{p}_j = P(\text{delay}_j > T)$ be the calibrated probability that job j will violate its SLO. The system applies action a if

$$\hat{p}_j \;\geq\; \varepsilon_a, \tag{1}$$

where ε_a is a tier-specific probability threshold.

3.2 Intervention Tiers

Table 1 enumerates the four tiers, their decision thresholds, and their calibration prerequisites.

The prerequisite column encodes the key insight: higher-stakes actions demand better-calibrated models. Tier 1 (Annotate) is informational—10 pp of miscalibration is still directionally correct (ECE $\leq$ 0.10). Tier 2 (Warn) interrupts the user's workflow, so excessive false warnings erode trust (ECE $\leq$ 0.07). Tier 3 (Suggest) implies high confidence in delay occurrence (ECE $\leq$ 0.05). Tier 4 (Gate) blocks submissions—users abandon systems where gating is wrong more than $\sim$ 3% of the time (ECE $\leq$ 0.03).

3.3 Graceful Degradation

When model quality degrades (ECE rises above a tier's prerequisite), the system automatically disables that tier and falls back to the highest tier still qualified. A model with ECE = 0.06 enables Tiers 1–2 but disables Tiers 3–4. This provides safety without total system shutdown and creates a natural feedback loop: degradation is visible in reduced functionality, motivating recalibration.

4 Submit-Time Capture Methodology

Correct decision rules require correct input features. We discovered a critical measurement error: periodic queue snapshots produce inverted feature correlations.

Table 2: Feature correlation: snapshot vs. submit-time capture.

Method	Correlation	Interpretation
Periodic snapshot (30 s)	$r = -0.27$	Inverted (wrong)
Submit-time capture (5 s)	$r = +0.44$	Correct

4.1 The Snapshot Timing Problem

Initial data collection captured queue state by periodic polling at 30 s intervals. Analysis revealed Pearson $r = -0.27$ between `pending_at_submit` and observed wait time: more pending jobs predicted shorter waits. The root cause is straightforward: queue state was captured at observation time, not at the moment of job submission. By the time a job is observed as "Pending," the queue has already drained.

4.2 Corrected Collection

We redesigned the collector to capture queue state at the exact moment of job submission: poll every 5 s, detect phase transitions (Pending → Running → Succeeded), and record queue state at each transition. Table 2 reports the corrected correlation.

This is not merely a data-quality issue—it is a decision-correctness issue. A model trained on snapshot-timed features would invert its recommendations: suggesting submission during high congestion and warning during low congestion. Submit-time capture is a prerequisite for correct decision rules, and is part of the VGAC instrumentation contract.

5 Experimental Validation

5.1 Dataset

Our EKS GPU cluster dataset contains 650 job-lifecycle records (582 usable after de-duplication) collected from an Amazon EKS cluster running on T4 and A10G GPUs. We validate the framework at two instrumentation levels to demonstrate that the tier system accommodates the full spectrum of deployment maturity.

Floor (2 features). The number of pending and running jobs at submit time, captured per the methodology of §4. This is the minimum viable predictor for a cluster with no resource instrumentation beyond queue depth.

Ceiling (17+ features): the full VGAC production feature set, comprising job attributes (CPU, memory, GPU, disk, RDMA, node selectors, affinity, tolerations), cluster state (pending ratio, node CPU and memory pressure, capacity metrics), temporal features (hour and day of week), and derived interaction terms. The ceiling model is trained on 11,982 GPU scheduling events across Kubernetes and Slurm workloads [2].

5.2 Floor Model Performance (2 Features)

For binary classification (long wait $>$ 120 s, base rate 54.1%) using only `pending_at_submit` and `running_at_submit`, Table 3 reports discrimination and calibration.

Table 3: Binary classification under the 2-feature floor: discrimination and calibration.

Metric	Value	Tier implication
AUROC	0.756	Adequate ranking
ECE	0.077	Qualifies Tiers 1–2
Precision	0.743	—
Recall	0.732	—
F1	0.738	—

Table 4: Tier qualification progression from floor to ceiling.

Configuration	AUROC	ECE	Tiers	Features
Floor (2 features)	0.756	0.077	1–2	2
Production VGAC (17+)	0.969	0.005	1–4	17+

5.3 Tier Qualification Assessment

This is the paper's central result. With ECE = 0.077, Tier 1 (Annotate) is qualified (ECE ≤ 0.10); Tier 2 (Warn) is marginal (ECE 0.077 exceeds 0.07 by 0.007 and would require recalibration to clear); Tiers 3–4 are not qualified. This honest assessment is invisible to AUROC-only evaluation. A system reporting AUROC = 0.756 might be deployed with gating enabled, but the ECE analysis reveals this would be premature. Examining the calibration bins, mid-range probabilities (0.35–0.52) show predicted-vs-actual gaps as large as 0.188—directly explaining why Tiers 3+ fail: estimates are least reliable in exactly the band where Suggest and Gate decisions are most consequential.

5.4 Floor to Ceiling: Feature Enrichment Unlocks Tiers

The graduated framework is designed to accommodate the full spectrum of instrumentation maturity. Table 4 shows the tier progression from the 2-feature floor to the 17+ feature production VGAC deployment.

With 17+ features and isotonic post-hoc calibration, the production VGAC model achieves ECE = 0.005—well below the Gate prerequisite of 0.03—qualifying for all four tiers including autonomous admission gating. The 12.7× improvement in ECE (0.077 → 0.005) demonstrates that the tier framework creates a concrete, measurable upgrade path: each investment in instrumentation (adding resource specifications, cluster-state capture, temporal features) directly translates into unlocked policy capabilities.

6 Integration Reference Implementations

We demonstrate integration via two production schedulers, both of which enforce tier qualification *at serve time* rather than only at deployment time, so degraded models automatically reduce their action scope.

Kubernetes. A mutating admission webhook intercepts Pod creation, extracts resource requests, queries the prediction service (< 10 ms), and annotates the Pod with the predicted probability and the qualified tier. The webhook checks the model's current rolling ECE before tier selection, so a model whose ECE has drifted above 0.05 will silently stop emitting Suggest and Gate annotations until recalibration restores qualification.

Slurm. A `job_submit` plugin attaches tier-qualified predictions to job metadata via the `Comment` field. Warn-level alerts are surfaced via `slurm_info` when the qualified tier and probability jointly warrant notification.

Both implementations follow the same pattern:

```
prob = predict(features);
tier = get_qualified_tier(prob, current_ece);
annotate(job, prob, tier).
```

The tier check is evaluated per request against the model's rolling ECE, ensuring that graceful degradation is dynamic rather than a build-time decision.

7 Discussion

The floor-to-ceiling validation (Table 4) demonstrates the framework's core value proposition: it accommodates deployments ranging from minimal instrumentation (2 features, Tiers 1–2) to full production observability (17+ features, all four tiers). The 2-feature floor proves that even minimal submit-time signal closes the reactive gap for advisory tiers. The 17+ feature ceiling proves that with proper instrumentation, calibration quality can support fully autonomous admission gating (ECE 0.005 ≪ 0.03 Gate prerequisite). Without calibration prerequisites, a team deploying the 2-feature model with AUROC 0.756 might enable gating; the framework prevents this by requiring each tier to be earned through demonstrated ECE. The upgrade path is explicit: add resource instrumentation → add cluster-state capture → add temporal features → watch tiers unlock.

Limitations. The ECE prerequisites in Table 1 are informed by operational intuition; formal cost-sensitive analysis could provide tighter bounds for specific operator cost structures. The 2-feature validation uses a single EKS cluster ($n = 650$); cross-cluster evaluation of tier qualification across heterogeneous schedulers is the subject of ongoing work and a companion benchmark [2].

8 Related Work

Decima [3] learns scheduling policies via reinforcement learning but does not address calibration or user-facing decision rules. Borg [6] and Firmament [7] optimize placement; our focus is the prediction-to-decision interface at submit time. Elkan [8] formalizes cost-sensitive classification, in which the optimal threshold depends on cost ratios; we extend this idea to graduated actions where each tier has a different cost profile and a different calibration requirement. Guo et al. [4] established ECE as a standard calibration metric; we operationalize ECE as a gate that constrains permissible actions.

9 Conclusion

We present VGAC, a full-stack GPU cluster observability platform whose predictive intelligence is calibration-gated through a decision-theoretic framework that translates calibrated queue-delay probabilities into scheduling actions. The key insight is that calibration quality determines permissible action severity: a model must *earn* each intervention tier through demonstrated ECE. We validate across the full instrumentation spectrum: a 2-feature floor model (ECE 0.077) honestly qualifies for annotation but not suggestion or gating, while the 17+ feature production VGAC model (ECE 0.005) qualifies for all four tiers including autonomous admission gating. This 12.7× ECE improvement from instrumentation investment creates a concrete upgrade path that is invisible to AUROC-only evaluation. The submit-time capture methodology ($r = -0.27 \rightarrow r = +0.44$) establishes that correct measurement is a prerequisite for correct decisions. Together with the companion reliability SLI/SLO framework [1], VGAC completes the path from calibrated prediction to safe scheduling policy.

Use of AI Assistance

Draft text of this paper was edited with the help of a large language model (Anthropic Claude). All results, tables, methodological choices, and interpretations were authored and verified by the human author; the language model did not generate experimental results. This disclosure follows ACM's policy on the use of generative AI in scholarly publications.

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